

Jihye Oh

Integrated M.S.–Ph.D. candidate · MAILAB, Department of Artificial Intelligence, Korea University

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Research Interests

Generative AI · LLM alignment and agentic AI · Medical AI

Education

Integrated M.S.–Ph.D., Department of Artificial Intelligence, Korea University Feb 2020 – present
MAILAB (Medical Artificial Intelligence Lab), advisor: Prof. Tae-Eui Kam

B.S. in Computer Science Engineering, Korea University of Technology and Education 2015 – 2020
(KOREATECH)

Experience

Winter research intern, Autonomous Driving Intelligence Research Laboratory, ETRI Jan – Feb 2020
Data preprocessing and semantic segmentation

Publications (26 total · 6 first or co-first author · 6 second author · * equal contribution)

International Journals (5)

1. A Hierarchical Reinforcement Learning Approach to Personalized Decision-Making for Brain Connectivity Segmentation
Chang-Hoon Ji, **Ji-Hye Oh**, Yu-Kyum Kang, Jun-Mo Kim, Suyeon Kwak, Ji-Wung Han, Sanghyeon Cho, Tae-Eui Kam
Expert Systems with Applications, 329, 132973, 2026
2. Source-Driven Enhanced Semantic Style Transfer for Addressing BCI Illiteracy in Subject-Independent EEG-Based Motor Imagery Classification
Ji-Wung Han, Dong-Hee Shin, Chang-Hoon Ji, Deok-Joong Lee, Jun-Mo Kim, WooHyeok Choi, **Ji-Hye Oh**, Sanghyeon Cho, Young-Han Son, Tae-Eui Kam
Applied Soft Computing, 2026
3. Graph-based Conditional Generative Adversarial Networks for Major Depressive Disorder Diagnosis with Synthetic Functional Brain Network Generation
Ji-Hye Oh, Deok-Joong Lee, Chang-Hoon Ji, Dong-Hee Shin, Ji-Wung Han, Young-Han Son, Tae-Eui Kam
IEEE Journal of Biomedical and Health Informatics, 28(3), 1504–1515, 2024
4. Spectral Graph Neural Network-based Multi-atlas Brain Network Fusion for Major Depressive Disorder Diagnosis
Deok-Joong Lee, Dong-Hee Shin, Young-Han Son, Ji-Wung Han, **Ji-Hye Oh**, Da-Hyun Kim, Ji-Hoon Jeong, Tae-Eui Kam
IEEE Journal of Biomedical and Health Informatics, 2024
5. Target-Aware Cross-Modality Unsupervised Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation
Bogyeong Kang, Hyeonyeong Nam, Myeongkyun Kang, Keun-Soo Heo, Minjoo Lim, **Ji-Hye Oh**, Tae-Eui Kam
Scientific Reports, 2024

International Conferences (16)

1. ProMoA-VPO: Progressive Multi-Objective Alignment for Video Prompt Optimization
Ji-Hye Oh*, Hyun Jung Lee*, Eunjung Jo*, Tae-Eui Kam
IEEE/CVF WACV 2027, Algorithms Track · accepted (Round 1)

2. PT-Diff: Brain Network Generation with Personalized Topology Guided Conditional Diffusion Transformer
Ji-Hye Oh, Chang-Hoon Ji, Su-Yeon Kwak, Yu-Kyum Kang, Ji-Wung Han, Jun-Mo Kim, Tae-Eui Kam
IEEE ICASSP 2026, Barcelona, pp. 7061–7065
3. Subject-Adaptive Multi-Subgraph Framework for Brain Graph Out-of-Distribution Generalization
Jaeyong Chang, **Ji-Hye Oh**, Suyeon Kwak, Tae-Eui Kam
IEEE SMC 2026, Bellevue
4. Reinforcement Learning with Multi-Objective Rewards for Functional Connectivity Augmentation
Chang-Hoon Ji*, Ji-Young Hwang*, **Ji-Hye Oh**, Yu-Kyum Kang, Su-Yeon Kwak, Ji-Wung Han, Tae-Eui Kam
IEEE ICASSP 2026, Barcelona
5. Adaptive Static-Dynamic Functional Connectivity Feature Fusion for Diagnosis of Mild Cognitive Impairment
Yu-Kyum Kang, Chang-Hoon Ji, **Ji-Hye Oh**, Su-Yeon Kwak, Ji-Wung Han, Tae-Eui Kam
IEEE ICASSP 2026, Barcelona
6. Brain Topology-Driven Graph Structure Learning for Functional Brain Network-Based Major Depressive Disorder Diagnosis
Suyeon Kwak, **Ji-Hye Oh**, Chang-Hoon Ji, Yu-Kyum Kang, Jaeyong Chang, Tae-Eui Kam
IEEE ISBI 2026, London
7. Agentic Brain: Agentic AI System for Autonomous Diagnosis of Functional Connectivity Network
Chang-Hoon Ji, **Ji-Hye Oh**, Yu-Kyum Kang, Suyeon Kwak, Ji-Wung Han, Jun-Mo Kim, Jiyoung Hwang, Tae-Eui Kam
ICEIC 2026, Macau
8. Adaptive Reward Weighting via Reinforcement Learning for Major Depressive Disorder Diagnosis
Jiyoung Hwang, Chang-Hoon Ji, **Ji-Hye Oh**, Yu-Kyum Kang, Suyeon Kwak, Ji-Wung Han, Tae-Eui Kam
ICEIC 2026, Macau
9. SeeEEG: Semantic-aware EEG-based Multi-Modal Retrieval-Augmented Generation for High-Fidelity Visual Brain Decoding
Jun-Mo Kim*, Woohyeok Choi*, Sang-Jun Park, Keun-Soo Heo, Young-Han Son, **Ji-Hye Oh**, Dong-Hee Shin, Tae-Eui Kam
ICCV Workshop (HiCV) 2025, Honolulu
10. Pre-to-Post Operative MRI Generation with Retrieval-based Visual In-Context Learning
Bogyeong Kang, Sang-Jun Park, Minjoo Lim, Myeongkyun Kang, Keun-Soo Heo, **Ji-Hye Oh**, Hyun Jung Lee, Tae-Eui Kam
MICCAI 2025, Daejeon, pp. 534–544
11. DART: Disease-aware Image-Text Alignment and Self-correcting Re-alignment for Trustworthy Radiology Report Generation
Sang-Jun Park*, Keun-Soo Heo*, Dong-Hee Shin, Young-Han Son, **Ji-Hye Oh**, Tae-Eui Kam
IEEE/CVF CVPR 2025, Nashville, pp. 15580–15589
12. Visual Decoding Using a Learnable Wavelet-Based Spatial-Spectral-Temporal EEG Embedding
Yebin Choi, Jun-Mo Kim, WooHyeok Choi, Chang-Hoon Ji, **Ji-Hye Oh**, Tae-Eui Kam
IEEE Winter Conference on Brain-Computer Interface 2025
13. Solving Blind Inverse Problem in Microscopy: Diffusion-Based Zero-Shot Isotropic Reconstruction
Hyunjung Lee, Eunjung Jo, Minjoo Lim, **Ji-Hye Oh**, Tae-Eui Kam
IEEE BIBM 2024, Lisbon
14. Image2SignalNet: Image-based Deep Learning Approach for Capturing Neuronal Signals from Calcium Imaging
Eunjung Jo, Dong-Hee Shin, **Ji-Hye Oh**, Sanghyeon Cho, Hyunjung Lee, Tae-Eui Kam
IEEE BIBM 2024, Lisbon
15. Integrating Nash Equilibrium with Reinforcement Learning for Improved Multi-Agent Coordination and Decision-Making
Chang-Hoon Ji, **Ji-Hye Oh**, Jun-Mo Kim, So-Yoen Park, Yu-Kyum Kang, Tae-Eui Kam
ICNGC 2024, Clark

16. A Comparative Study: Enhancing Conditional Generative Adversarial Networks for Functional Connectivity Synthesis in Major Depressive Disorder
Ji-Hye Oh, Eunjung Jo, Tae-Eui Kam
ICPRAI 2024, Jeju

Domestic Conferences (5)

1. Brain Network Generation with Conditional Latent Graph Diffusion
Ji-Hye Oh, Chang-Hoon Ji, Jun-Mo Kim, Jiyoung Hwang, Tae-Eui Kam
Conference on Electronics, Semiconductor, and AI 2025

2. Synthetic Augmentation of Functional Connectivity using R3GAN for Major Depressive Disorder Diagnosis
Jiyoung Hwang, Chang-Hoon Ji, **Ji-Hye Oh**, WooHyeok Choi, Jun-Mo Kim, Tae-Eui Kam
Conference on Electronics, Semiconductor, and AI 2025

3. Implicit and Explicit Domain Alignment for Cross-dataset Brain-Computer Interface
Jun-Mo Kim, Chang-Hoon Ji, **Ji-Hye Oh**, So-Yoen Park, WooHyeok Choi, Jiyoung Hwang, Tae-Eui Kam
Conference on Electronics, Semiconductor, and AI 2025

4. Towards Reliable Policy Convergence in Multi-Agent Systems with Equilibrium-Based Deep Reinforcement Learning
Chang-Hoon Ji, **Ji-Hye Oh**, Jun-Mo Kim, Tae-Eui Kam
Conference on Electronics, Semiconductor, and AI 2024

5. Dynamic Connectivity Pattern Generation Using Time-Series GANs for Autism Spectrum Disorder Diagnosis
Ji-Hye Oh*, Chang-Hoon Ji*, Tae-Eui Kam
Korean Institute of Next Generation Computing, Spring Conference 2024 · Best Oral Paper Award

Review Experience

IEEE Journal of Biomedical and Health Informatics, IEEE Transactions on Medical Imaging, Expert Systems with Applications, ICLR, IEEE BIBM, IEEE ICASSP

Patent

Brain Disease Diagnosis Device and Method Using Graph-Based Conditional Adversarial Generative Networks 2024
Korean patent application 10-2024-0108708

Award

Best Oral Paper Award, Korean Institute of Next Generation Computing Spring Conference 2024